

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 13

2024

1. Provide an example that shows that complete, bounded metric spaces need not be compact.
2. Use Theorem 10.16 to prove the following.

Corollary Suppose that $\{C_j : j \in J\}$ is a collection of connected subsets of a metric space X , and B is also a connected subset of X . Further, suppose that, for every $j \in J$, we have $C_j \cap B \neq \emptyset$. Then $B \cup \bigcup_{j \in J} C_j$ is connected.

3. **Definition** Let X be a connected metric space. We call $c \in X$ a *cut point* of X if $X - \{c\}$ is disconnected.

Theorem Let $f : X \rightarrow Y$ be a homeomorphism of metric spaces and X be a connected space. If c is a cut point of X , then $f(c)$ is a cut point of Y .

Proof Recall that $f(\bigcup_i A_i) = \bigcup_i f(A_i)$ and $f(\bigcap_i A_i) \subseteq \bigcap_i f(A_i)$ for all collections $\{A_i : i \in I\}$ of subsets of X . (1)

If f is bijective, then we also have $f(A_1 \cap A_2) = f(A_1) \cap f(A_2)$ for all subsets A_1 and A_2 of X . (2)

Write $B = X - \{c\}$, where c is a cut point of X . (3)

There exist G_1 and G_2 open in X , such that $B = (B \cap G_1) \cup (B \cap G_2)$ and $B \cap G_1$ and $B \cap G_2$ are disjoint and non-empty. (4)

Apply f to get $f(B) = (f(B) \cap f(G_1)) \cup (f(B) \cap f(G_2))$ and $f(B) \cap f(G_1)$ and $f(B) \cap f(G_2)$ disjoint and non-empty, and $f(G_1)$ and $f(G_2)$ open. (5)

This shows that $f(B)$ is disconnected. (6)

Further, $f(B) = f(X - \{c\}) = Y - \{f(c)\}$. (7)

So $f(c)$ is a cut point of Y . (8)

- (a) Prove the claim about intersections in line (1).
- (b) Prove the claim in line (2). In fact, you do not need the full strength of “bijection.” What do you actually use?
- (c) Why do the sets G_1 and G_2 with the properties listed in line (4) exist?
- (d) Why is $f(G_1)$ open, as claimed in line (5)?
- (e) Why is $f(B) \cap f(G_1) \neq \emptyset$ as claimed in line (5)?
- (f) Prove the second equality in line (7).
- (g) To get from line (7) to line (8) we technically need the fact that Y is connected. How do you know this?

4. Let (A_n) be a sequence of connected subsets of a metric space (X, d) such that $A_n \cap A_{n+1} \neq \emptyset$ for each $n \in \mathbb{N}$. Show that $\cup_{n \in \mathbb{N}} A_n$ is connected.
5. Show that if A is a connected subset of a metric space (X, d) , then there is a component C of X such that $A \subseteq C$.
6. Prove that a clopen connected subset of a metric space (X, d) is a component of X .